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Extracting key insights from earnings call transcript via information-theoretic contrastive learning

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ABSTRACT

Earnings conference calls provide critical insights into a company's financial health, future outlook, and strategic direction. Traditionally, analysts manually analyze these lengthy transcripts to extract key information, a process that is both time-consuming and prone to bias and error. To address this, text mining tools, particularly extractive summarization, are increasingly being used to automatically extract key insights, aiming to standardize the analysis process and improve efficiency. Extractive summarization automates the selection of the most informative sentences, offering a promising solution for transcript analysis. However, existing extractive summarization techniques face several challenges, such as the lack of labeled training data, difficulties in incorporating domain-specific knowledge, and inefficiencies in handling large-scale datasets. In this work, we introduce ECT-SKIE, an information-theoretic, self-supervised approach for extracting key insights from earnings call transcripts. We leverage variational information bottleneck theory to extract insights in parallel, significantly accelerating the process. In addition, we propose a structure-aware contrastive learning strategy that enables model training without the need for labeled data. We further develop a novel container-based key sentence extractor to alleviate sentence redundancy. Using a large-scale dataset of U.S. market earnings call transcripts, we evaluate our method against nine representative baselines across three downstream tasks. Experimental results show that ECT-SKIE can consistently extract high-quality key sentences. The code is publicly available at: https://github.com/MongoTap/ECT-SKIE.

1. Introduction

Earnings calls are routine teleconferences in which the management of a public company discusses its financial results for a specific reporting period. These calls provide valuable insights into a company's performance, future prospects, and strategic initiatives that are often not fully captured in financial statements alone (Blythe, 2023; Rennekamp, Sethuraman, & Steenhoven, 2022; S&P Global, 2023). Traditionally, industry professionals meticulously review earnings call transcripts to extract key information and use these insights to inform decisions in areas such as portfolio management (Doran, Peterson, & Price, 2012) and risk assessment (Sang & Bao, 2022b). However, this analysis process demands significant expertise and effort (Group, 2023). A typical earnings call transcript can span thousands of words, often covering both prepared remarks and an hour-long Q&A session, making it a time-consuming

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task even for seasoned financial experts: analyzing a single transcript can take several hours or an entire workday (Abbas, 2023). Furthermore, manual analysis is susceptible to human bias and error. To address these challenges, there is a growing trend toward using text mining tools to automatically extract key insights from earnings call transcripts, aiming to standardize the analysis process and improve efficiency (Huang, Lehavy, Zang, & Zheng, 2018; Medya, Rasoolinejad, Yang, & Uzzi, 2022; Yang, Zhang, & Fan, 2022).

The extraction of key insights from earnings call transcripts involves various analytical techniques, such as keyword extraction (Bloom, Hassan, Kalyani, Lerner, & Tahoun, 2021; Li, Mai, Shen, & Yan, 2021), sentiment analysis (Koval, Andrews, & Yan, 2023; Qin & Yang, 2019), and topic modeling (Huang et al., 2018; Yang et al., 2022). Among them, keyword extraction identifies important words or phrases, sentiment analysis gauges the emotions and opinions of management, and topic modeling distills central themes from the discussion. While these methods can efficiently pinpoint specific elements within the text, they often provide a fragmented view of the overall context. In contrast, key sentence extraction retains the context and integrity of the crucial information, offering a more comprehensive perspective that better aligns with the practical needs of investors and market analysts (Jacobs & Hoste, 2022).

A common approach to extracting key sentences is through extractive summarization methods (Erkan & Radev, 2004; Filippova, Surdeanu, Ciaramita, & Zaragoza, 2009; Leidner, 2020; Mihalcea & Tarau, 2004; Mukherjee et al., 2022; Nugent, Gkotsis, & Leidner, 2023; Padmakumar & He, 2021). However, directly applying these techniques to earnings call transcripts presents several challenges. First, these transcripts typically lack annotations, and manual labeling is both time-consuming and prone to bias, making supervised summarization methods impractical for this task. Second, most unsupervised or self-supervised algorithms are task-agnostic, limiting their ability to incorporate domain-specific knowledge when assigning importance to sentences. Finally, many advanced extractive approaches (Padmakumar & He, 2021; West, Holtzman, Buys, & Choi, 2019) rely on Point-wise Mutual Information (PMI) to measure sentence relationships, which leads to a computational complexity of at least $\mathcal{O}(N^2)$, severely impacting processing efficiency.

An alternative solution is to leverage large language models (LLMs) such as Llama, Gemini, and ChatGPT for key insight extraction (Lee, Stevens, Han, & Song, 2024; Zhao et al., 2024). These models offer the advantage of being pre-trained on vast amounts of data, allowing them to generate contextually relevant summaries without the need for task-specific training. However, several limitations hinder their effectiveness in extracting key sentences from earnings call transcripts. Firstly, LLMs require substantial computational resources, making them less efficient for processing large-scale earnings call transcripts compared to more lightweight extractive methods. Secondly, as an autoregressive model, LLMs generate summaries rather than directly extracting key sentences from the original text. This means the output lacks interpretability, as it is difficult to trace back the specific sentences or content in the transcript that contributed to the generated summary, leading to a lack of transparency compared to extractive methods like TextRank, where selected sentences are explicitly taken from the text (Zhou, Ringeval, & Portet, 2023). Lastly, although LLMs are highly versatile, they may struggle with domain-specific content in earnings call transcripts, as they often lack the specialized knowledge required to accurately weigh the importance of key information in niche financial contexts (Zhang, Liu, & Zhang, 2023).

In this work, we propose ECT-SKIE (**E**arnings **C**all **T**ranscript via **S**tructure-Aware **K**ey **I**nsight **E**xtraction), an information-theoretic framework designed to extract key sentences in a self-supervised manner. We first introduce a variational information bottleneck technique to compress information in parallel, filtering out irrelevant sentences while preserving the most critical ones. Then, we also propose a structure-aware contrastive learning strategy that leverages the transcript's inherent structure to provide label-free training signals. To address the issue of sentence redundancy that can arise from parallel extraction – where semantically similar sentences may be considered equally important – we develop a novel container-based key sentence extractor to effectively reduce redundancy. We collect 41,511 earnings call transcripts from platforms such as SeekingAlpha and Thomson Reuters StreetEvents and evaluate our method across three common downstream tasks: risk forecasting, transcript retrieval, and firm discovery, comparing it to nine representative baselines. The experimental results demonstrate the high quality of the extracted key sentences: (1) Using a pre-trained language model (BERT Kenton & Toutanova, 2019) to encode the key sentences into dense vectors, followed by a simple three-layer Multi-Layer Perceptron (MLP), ECT-SKIE outperforms state-of-the-art methods designed specifically for risk forecasting; (2) Applying the same dense vectors for transcript retrieval results in superior performance compared to existing extractive summarization methods; (3) The transcript embeddings produced by ECT-SKIE effectively identify economically related companies.

In summary, our contributions are three-fold:

- We introduce a novel approach for extracting key insights from earnings call transcripts. The proposed approach incorporates variational information bottleneck theory with structure-aware contrastive learning, enabling the extraction of key insights in a self-supervised and parallel manner. This allows us to train models without using any golden labels and achieves a speedup of 500 times compared to existing PMI-based extractive summarization methods.
- We propose a novel container-based key sentence extractor to address the redundancy issue, thereby ensuring the extracted insights are concise and informative.
- Extensive experiments on a large-scale dataset validate the effectiveness of ECT-SKIE in extracting key insights.

The rest of the paper is structured as follows. Section 2 reviews related work, while Section 3 covers the background and outlines the research question. In Section 4, we introduce the proposed ECT-SKIE framework in detail. Section 5 presents an evaluation of our method across three representative tasks, comparing it against nine advanced baselines. Finally, Section 6 discusses the limitations of the study and suggests potential directions for future research.

2. Related works

We review three main categories of closely related literature: (1) unsupervised extractive summarization, (2) information bottleneck theory in NLP, and (3) financial analysis based on earnings conference calls.

2.1. Unsupervised extractive summarization

Unsupervised extractive summarization is a natural language processing (NLP) technique that generates summaries by selecting the most important sentences from a text without requiring labeled data. Key approaches include LexRank (Erkan & Radev, 2004), TextRank (Mihalcea & Tarau, 2004), PACSUM (Zheng & Lapata, 2019), TED (Yang, Zhu et al., 2020), STAS (Xu, Zhang, Wu, Wei, & Zhou, 2020), and PMI (Padmakumar & He, 2021). LexRank measures the importance of sentences in a document by using eigenvector centrality on a graph where edges represent cosine similarity between sentences. TextRank represents the text as a graph with sentences as nodes and edges based on semantic similarity, using the PageRank algorithm to rank and select top sentences. PACSUM uses BERT to calculate sentence similarity and constructs a text graph with directed edges. Sentences with high centrality are regarded as important. TED pre-trains a model on millions of unlabeled news articles by exploiting the lead bias, then fine-tunes it using theme modeling and a denoising autoencoder to improve the semantic relevance and quality of the generated summaries. STAS pre-trains a hierarchical transformer model with masked sentence prediction and shuffling unsupervised objectives, then ranks sentences based on attention scores. PMI leverages pointwise mutual information (PMI) between sentences to evaluate their significance.

Existing unsupervised or self-supervised summarization algorithms are task-agnostic and inefficient. In contrast, our method leverages the inherent structure of earnings call transcripts and enable parallel extraction, which is more effective and efficient than existing methods.

2.2. Information bottleneck theory in NLP

The Information Bottleneck (IB) theory (Tishby & Zaslavsky, 2015) has been influential in understanding the compression and denoising processes in deep neural networks, particularly in the context of NLP. The core concept of IB theory is that a model should ideally compress the input data while preserving the most relevant information for the task at hand. For example, West et al. (2019) apply the IB theory to guide the summarization process by identifying and retaining information relevant for predicting the next sentence in a document, thus compressing the input while preserving its essential content. Bang, Xie, Lee, Wu, and Xing (2021) compress the input information while retaining its relevance to the downstream task, which was achieved by selecting informative words from sentences by minimizing the variational lower bound of IB.

In this work, we demonstrate that directly leveraging IB theory to extract key sentences suffers from sentence redundancy. To address this, we propose a novel container-based key sentence extractor, which mitigates redundancy and enhances the conciseness of the extracted insights.

2.3. Financial analysis based on earnings conference calls

Earnings conference calls provide valuable insights into a company's recent performance, strategic direction, and future outlook, which are key factors influencing market sentiment and investment decisions (Brockman, Li, & Price, 2015; Chen, Liao, Chen, Kang, & Lin, 2023; Hajek & Munk, 2024). Early efforts to quantify the informational content of financial reports rely on dictionaries to assess word polarity, such as the Harvard General Inquirer IV-4 (HGI) dictionary (Li et al., 2010). Loughran and McDonald (2011) develop a finance-specific sentiment dictionary, showing that many words perceived as negative in general contexts have different connotations within financial texts.

Recent advancements in deep learning lead to more sophisticated methods for analyzing earnings conference calls. For instance, Qin and Yang (2019) propose a deep multi-modal regression model that integrates both textual and audio data from conference calls to predict short-term stock volatility. Sawhney et al. (2020) extend this by incorporating stock correlation networks and financial features using graph networks for joint predictions. Koval et al. (2023) demonstrate that it is feasible to forecast a company's future earnings surprises with reasonable accuracy using only the textual content of their conference calls. Later on, Jha, Qian, Weber, and Yang (2024) utilize large language models (LLMs) to extract firms' future investment expectations from earnings call transcripts, developing an Investment Score that effectively predicts future capital expenditures.

3. Preliminaries

We first provide a brief introduction to the earnings conference call. Then we define and discuss the key insights (sentence) extraction task.

3.1. Earnings call

As shown in Fig. 1, an earnings call consists of a *Presentation* section and a *Q&A* section. In *Presentation*, executives such as CEOs or CFOs read forward-looking statements and provide their opinions and interpretations of the firm's quarterly performance.

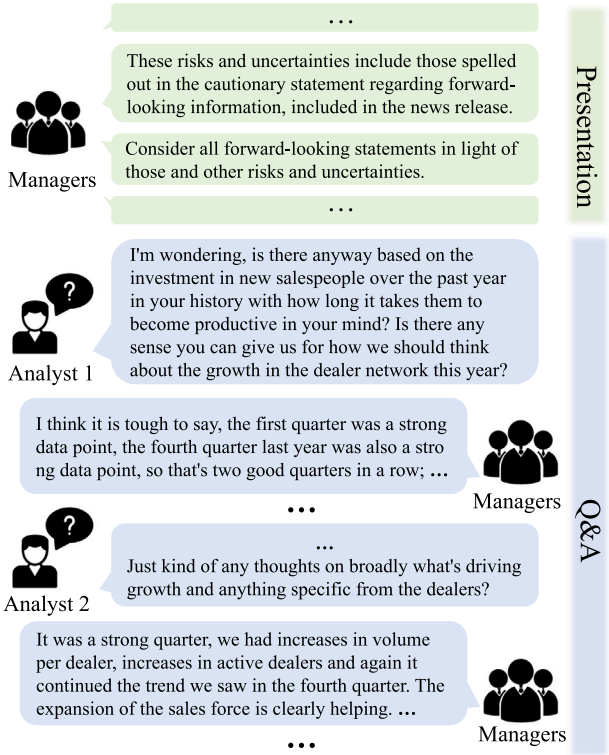

Fig. 1. Structure of an earnings conference call. After the company's management presents their report, multiple Q&A rounds take place between the analysts and the management team.

In the *Q&A* section, analysts can ask follow-up questions, request clarification on the information presented earlier, or solicit new information not disclosed in the Presentation section. These calls generally last about an hour, with transcripts often containing thousands of words. In this work, we collect large-scale earnings call transcripts of U.S. firms from Seekingalpha and Thomson Reuters StreetEvents. Then, we extract the transcripts in four fiscal years (2015–2018) for training, validating, and testing and present the statistics of them in Table 1.

3.2. Task formulation

Given an earnings call transcript containing totally N sentences, our objective is to train a neural network (NN) to extract key sentences sufficient for various downstream tasks. Formally, let $h_\theta(\cdot)$ denote a shared representation function that maps the input data to a representation in a higher-dimensional space, $g_i(\cdot)$ denote a task-specific function mapping the high-dimensional representation to the output space of task $\mathcal{T}$, and $\mathcal{L}(\cdot)$ denote a loss function that measures the discrepancy between the predictions and the ground-truth. The core of our method is to find a key sentence set $\kappa(X)$ given a transcript X that satisfies:

$$\mathbb{E}_{\mathcal{T}}\left[\mathcal{L}(g(h_\theta(\kappa(X))), Y)\right] \approx \mathbb{E}_{\mathcal{T}}\left[\mathcal{L}(g(h_\theta(X)), Y)\right]. \tag{1}$$

We seek to optimize the following objective function:

$$\kappa_\psi(X)^* = \underset{g_1, g_2, \ldots, g_{|\mathcal{T}|}}{\operatorname{argmin}} \sum_{k=1}^{|\mathcal{T}|} \gamma_k \mathcal{L}_k(g_k(h_\theta(X)), Y),$$

where hyperparameters $\{\gamma_k\}_{k=1}^{|\mathcal{T}|}$ control the intensity of each task. Since manual labeling (finding key sentences) is time-consuming and prone to error and bias, in the context of our task, we work exclusively with the collected earnings call transcripts and have no access to any golden labels when extracting key sentences. Moreover, given that we lack ground-truth labels for the assessment of key sentences, we need to evaluate the model on several downstream tasks, which also position our model, ECT-SKIE, as a representation learning approach. In this procedure, we first extract the key sentences, followed by the use of arbitrary off-the-shelf language models such as BERT to generate a dense vector per transcript.

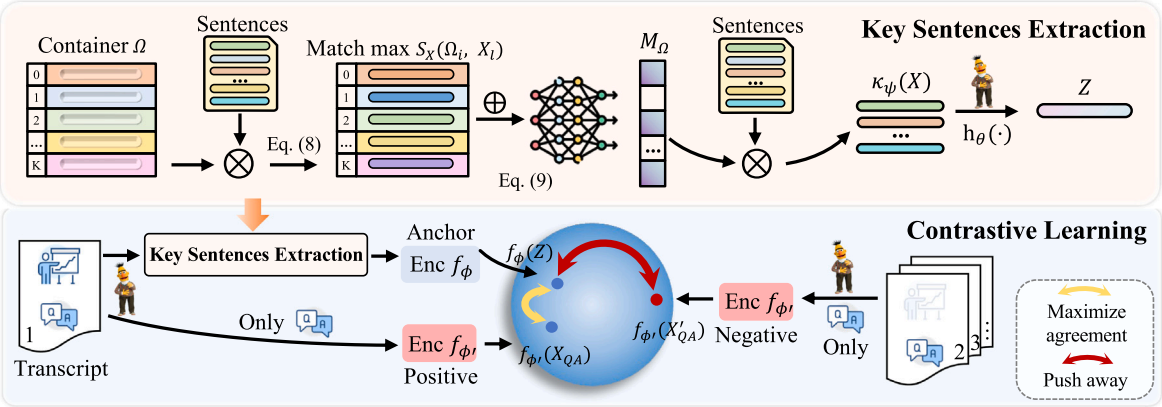

Fig. 2. Overall workflow of ECT-SKIE. Earnings call transcripts are encoded and fed into the key sentences extractor, generating a mask denoted as M_Ω and the corresponding transcript representation Z. ECT-SKIE is optimized via InfoNCE objective, in which Z is viewed as the anchor. The representation of the Q&A section is encoded as the positive sample, while representation of the Q&A section from other transcripts in the batch are treated as negative samples.

4. Methodology

In the following, we first introduce variational information bottleneck (IB) technology to enable parallel extraction of key insights from a lengthy transcript (Section 4.1). Then, we propose a structure-aware contrastive learning method and illustrate how to extract key sentences in a self-supervised manner (Section 4.2). The overall workflow of ECT-SKIE is shown in Fig. 2.

4.1. Key insights extraction

A direct solution for extracting key sentences is to devise an NN $f_\psi(X) : \mathbb{R}^{N\times d} \to \mathbb{R}^N$ that generates a binary mask $M \in \mathbb{R}^N$ for each transcript with N sentences – ψ denotes the trainable parameters of the extractor. Accordingly, we can implement a key sentence extractor via:

$$\kappa_\psi(X) = f_\psi(X) \odot X, \tag{2}$$

where $\kappa_\psi(X)$ is the selected sentence set by NN, and $\odot$ denotes the key sentence selector, i.e., we put sentences X_i into $\kappa_\psi(X)$ if M_i equals 1. Accordingly, we can optimize the model via:

$$\psi^* = \arg\min_\psi \mathbb{E}_\mathcal{T}\left[\mathcal{L}(g(h_\theta(\kappa_\psi(X))), Y)\right]. \tag{3}$$

However, directly optimizing Eq. (3) may also extract those unimportant sentences. To address this issue, we introduce Information Bottleneck (IB) theory (Tishby & Zaslavsky, 2015) and optimize the model via:

$$\psi^* = \arg\min_\psi \mathbb{E}_\mathcal{T}\left[\mathcal{L}(g(h_\theta(\kappa_\psi(X))), Y)\right] + \beta\ I(X;\kappa_\psi(X)), \tag{4}$$

where β is a weight hyper-parameter that balances the trade-off between compression ratio and information sufficiency. The motivation behind Eq. (4) is two-fold. First, we encourage the NN to extract sufficient knowledge for downstream tasks. Second, we also need to maintain the conciseness of $\kappa_\psi(X)$. In other words, we encourage the model to "select" really key sentences and M to be sparse.

One approach to estimating the second term $I(X, \kappa_\psi(X))$ involves using computationally expensive estimation techniques, such as the Donsker–Varadhan representation of mutual information (Belghazi et al., 2018). Inspired by prior studies (Bang et al., 2021; Kim, Kim, Woo, & Kim, 2021), we employ a more efficient way by exploring a variational approximation of the second term as:

$$I(X, \kappa_\psi(X)) \le \mathbb{E}\left[D_{KL}(\mathbb{P}_\psi(M|X), r(M))\right], \tag{5}$$

where $r(M)$ is the prior distribution of the mask M. The derivation is provided below, where we replace the key sentence set $\kappa(X)$ with X_s to simplify the formulas.

Firstly, we show that $\mathrm{I}(X, X_s) \le \mathrm{I}(X, M_s) + C$ where C is a constant C and then use the lower bound for $-\mathrm{I}(X, M_s) - C$ as a lower bound for $-\mathrm{I}(X, X_s)$. From the Markov Chain $X \to (X, M_s) \to X_s$, we have $\mathrm{I}(X, X_s) \le \mathrm{I}(X, (X, M_s))$. According to the chain rule for mutual information, $\mathrm{I}(X, (X, M_s)) = \mathrm{I}(X, M_s) + \mathrm{I}(X, X|M_s)$, where $\mathrm{I}(X, X|M_s) = \mathrm{H}(X|M_s) + \mathrm{H}(X|M_s) - \mathrm{H}(X, X|M_s)$. Further, since $\mathrm{H}(X|M_s) \le \mathrm{H}(X)$, we have

$$\mathrm{I}(X, X_s) \le \mathrm{I}(X, M_s) + \mathrm{H}(X) \tag{6}$$

where the entropy $\mathrm{H}(X)$ of input is a constant. We then approximate $p(M_s)$ using $r(M_s)$. Given that the KL divergence is always positive, we have

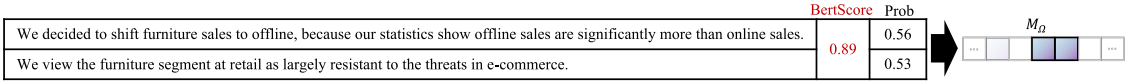

Fig. 3. Illustrative cases of redundancy issue. Due to the characteristics of shared neurons, two sentences with similar meanings get similar scores. As such, maintaining only one sentence is enough.

$$\begin{aligned}\mathbb{E}_{(X,M_s)\sim p(X,M_s)}[\log p(M_s)] &= \mathbb{E}_{M_s\sim p(M_s)}[\log p(M_s)]\\ &\geq \mathbb{E}_{M_s\sim p(M_s)}[\log r(M_s)]\\ &= \mathbb{E}_{(X,M_s)\sim p(X,M_s)}[\log r(M_s)].\end{aligned} \tag{7}$$

From (6) and (7), we have

$$\begin{aligned}\mathrm{I}(X,X_s) \leq \mathrm{I}(X,M_s) + C &= \mathbb{E}_{(X,M_s)\sim p(X,M_s)}\left[\log \frac{p(M_s \mid X)}{p(M_s)}\right] + C\\ &\leq \mathbb{E}_{(X,M_s)\sim p(X,M_s)}\left[\log \frac{p(M_s \mid X)}{r(M_s)}\right] + C\\ &= \mathbb{E}_{X\sim p(X)} D_{\mathrm{KL}}(p(M_s \mid X), r(M_s)) + C.\end{aligned} \tag{8}$$

If we use a NN $\phi_\psi(X)$ to approximate $p(M_s|X)$, then we have:

$$I(X,\kappa_\psi(X)) \leq \mathbb{E}\left[D_{KL}(p_\psi(M_s|X), r(M_s))\right], \tag{9}$$

which means that we can minimize $I(X,\kappa_\psi(X))$ through minimizing $\mathbb{E}\left[D_{KL}(p_\psi(M_s|X), r(M_s))\right]$.

Pitfalls in optimizing Eq. (5)**.** Typically, for the NN $f_\psi(X)$ and the input sentences $X \in \mathbb{R}^{N\times d}$, the binary mask $M \in \mathbb{R}^N$ is the output of $f_\psi(X)$ where 1 indicates being selected as a key sentence and 0 otherwise. Although we have applied the IB strategy to force the model to generate concise explanations, the above key sentence extractor still suffers from the redundancy issue in practice. As shown in Fig. 3, if two sentences are semantically similar (and important), both will be selected as key sentences.

Redundancy-aware key sentence extractor. To address the sentence redundancy, we propose a simple yet effective solution by adding a container with K slots $\Omega \in \mathbb{R}^{K\times d}$ ($K \ll N$). Instead of determining the importance of sentences directly, we alternatively consider which slot in the container is important. Specifically, the redundancy-aware key sentence extractor consists of the following two steps:

(1) Bind candidate sentences into the container. We calculate the score between sentence X_l and slot Ω_i via:

$$S_X(\Omega_i, X_l) = \frac{\exp(\mathrm{sim}(\Omega_i, X_l))}{\sum_{j=1}^{N}\exp(\mathrm{sim}(\Omega_i, X_j))}, \tag{10}$$

where $\mathrm{sim}(\Omega_i, X_l)$ is cosine similarity. The slot Ω_i will bind with the sentence S with the highest score:

$$\Omega_i(S) = \arg\max_{S=X_l} S_X(\Omega_i, X_l), \quad \text{for } l = 1,\dots,N. \tag{11}$$

As a result, each slot will hold one sentence and a sentence may be associated with multiple slots. Please note that the *argmax* operation in Eq. (11) does not support backpropagation. To approximate the *argmax* operation, we use the generalized Gumbel-Softmax trick (Jang, Gu, & Poole, 2017).

(2) Select top-ranked slots. After associating candidate sentences with the container, our next goal is to select the top-N_s ranked slots, thereby extracting at most N_s key sentences. To achieve this, we first create a transcript-aware slot representation by combining the slot representation with its corresponding sentence vector. Next, we concatenate the representation of the slot and the sentence it belongs to. We then employ $\mathrm{MLP}(\Omega_i \oplus X_j) : \mathbb{R}^{2d} \to \mathbb{R}$ to generate the slot scores.

$$S_\Omega(\Omega_i, S_i) = \frac{\exp(\mathrm{MLP}(\Omega_i \oplus S_i))}{\sum_{j=1}^{K}\exp(\mathrm{MLP}(\Omega_j \oplus S_j))}. \tag{12}$$

We select the top-N_s slots and extract their corresponding sentences to construct $\kappa_\psi(X)$. The new training objective then becomes:

$$\psi^* = \arg\min_\psi \mathbb{E}_{\mathcal{T}}\left[\mathcal{L}(g(h_\theta(\kappa_\psi(X))), Y)\right] + \beta\, \mathbb{E}\left[D_{KL}(\mathbb{P}_\psi(M_\Omega|X), r(M_\Omega))\right]. \tag{13}$$

Consequently, we rewrite the mask M for sentences in Eq. (5) as M_Ω for slots. These modifications enable the model to effectively address the redundancy issue: When two sentences have similar semantics, they are likely to belong to the same slot with high probability. The *argmax* operation ensures that only one sentence is retained among a group of sentences with similar semantics.

Notably, we do not require any additional loss term to ensure that different slots focus on distinct topics. If all slots were to concentrate on the same topic, the sentence selection process would degenerate into uniform sampling, resulting in a high loss for Eq. (13). Our experiments demonstrate that the proposed container performs well in practice (cf. Section 5.5).

Table 1
Descriptive statistics of earnings call transcripts.

Year	2015	2016	2017	2018
# transcripts	10,168	9765	10,431	11,147
# firms	3,398	3427	3,451	3,616
avg. # tokens in Presentation	3,207	3172	3,191	3,199
avg. # tokens in Q&A	4,347	4197	4,222	4,245

4.2. Self-supervised representation learning

Unfortunately, direct optimizing Eq. (13) is usually impractical due to the limited availability of transcripts and downstream data (Mukherjee et al., 2022). Therefore, we have to design a new training objective that is independent of the downstream data. As previously mentioned, key sentences in earnings call transcripts provide crucial information about a company's financial performance and future prospects. In the Q&A section, for example, analysts often ask follow-up questions and request clarification on the presented information. This interaction highlights what investors consider important, as the topics they inquire about typically relate to vital company information (Miwa, 2023). This observation presents an opportunity to design a surrogate supervision signal by leveraging Q&A sections to implement self-supervised learning. Specifically, we can replace the first term in Eq. (13) with:

$$\arg\min_{\psi} \mathbb{E}_{\mathcal{T}}\left[\mathcal{L}(g(h_\theta(\kappa_\psi(X))), Y)\right] \rightarrow \arg\max_{\psi} I(\kappa_\psi(X); X_{QA}) = \mathbb{E}\left[\log \frac{p\left(\kappa_\psi(X), X_{QA}\right)}{p\left(\kappa_\psi(X)\right) p\left(X_{QA}\right)}\right], \tag{14}$$

where X_{QA} is the set of sentences in the Q&A section. Considering the intractability of the computation of Eq. (14), we maximize the mutual information between X_s and X_{QA} by optimizing a contrastive loss – InfoNCE (Oord, Li, & Vinyals, 2018). Specifically, the mutual information between X_s and X_{QA} is defined as:

$$I(\kappa_\psi(X), X_{QA}) = \sum_{\kappa_\psi(X), X_{QA}} p(\kappa_\psi(X), X_{QA}) \log \frac{p(\kappa_\psi(X)|X_{QA})}{p(X_s)}.$$

We encode the extracted N_s key sentences $\kappa_\psi(X)$ using BERT, followed by a mean pooling operation to compress multiple representations into a single vector. For the Q&A section, which has inherent structural information divisible into multiple rounds, we feed each round into BERT and use the mean vector of all rounds as the representation of the entire Q&A section. We then employ infoNCE (Chen, Kornblith, Norouzi, & Hinton, 2020) to estimate the lower bound of Eq. (14):

$$\mathcal{L}_{\text{NCE}} = -\log \frac{\exp(f_\phi(h_\theta(\kappa_\psi(X))) \cdot f_{\phi'}(h_\theta(X_{QA})))}{\sum_{\mathcal{B}} \mathbb{1}_{X'_{QA} \notin X} \exp(f_\phi(h_\theta(\kappa_\psi(X))) \cdot f_{\phi'}(h_\theta(X'_{QA})))}, \tag{15}$$

where f_ϕ and f'_ϕ are linear layers, $\mathcal{B}$ denotes the batch size, and $\mathbb{1}$ is an indicator function. Note that, for conciseness, we omit the symbols for the pooling operations and temperature hyper-parameter tuning the balance between positive and negative samples in Eq. (15). Finally, the task-agnostic self-supervised training objective is defined as:

$$\mathcal{L} = \mathcal{L}_{\text{NCE}} + \beta\ D_{KL}(p_\psi(M^*_\Omega|X), r(M_\Omega)). \tag{16}$$

Given a transcript with N sentences, we select N_s sentences with a pre-defined ratio $\alpha = \frac{N_s}{N}$ and use a uniform distribution for $r(M_\Omega)$.

5. Experimental results

In this study, we conduct comprehensive evaluations to evaluate the effectiveness of ECT-SKIE on three tasks: risk prediction, transcript retrieval, and firm discovery.

5.1. Datasets

We collect a large-scale earnings transcript dataset of U.S. firms for four fiscal years (2015–2018). The transcripts are available at the SeekingAlpha[2] or databases such as Thomson Reuters StreetEvents.[3] The scale and time period of the dataset are similar to those used in prior literature (Sang & Bao, 2022a; Yang et al., 2022; Ye, Qin, & Xu, 2020). The breakdown of the dataset is presented in Table 1. The descriptive statistics show that earnings call transcripts are typically very long. On average, there are more than 3000 words in the Presentation section and more than 4000 words in the Q&A section. In addition to the earnings conference call transcripts, we obtain each company's daily stock prices from the CRSP database.[4] Since each earnings transcript is associated

[2] https://seekingalpha.com.
[3] https://www.streetevents.com.
[4] https://crsp.org.

with a date when the call took place, we calculate the risk τ-day after the earnings call, where $\tau \in \{3, 7, 10, 15, 20, 60\}$ indicates different forecasting horizons, i.e., daily to weekly risk ($\tau = 3, 7$), weekly to monthly risk ($\tau = 10, 15, 20$), and quarterly risk ($\tau = 60$). Following Yang, Qin, Fan, and Zhang (2023), we choose data from 2015 and 2016 as the training set and use data from 2017 as the validation set. Data from 2018 serves as the testing set. This chronologically divided dataset setup can prevent look-ahead bias.

5.2. Experiment 1: Overall Performance of ECT-SKIE on Risk Prediction

In this experiment, we evaluate the overall performance of the proposed method ECT-SKIE on the risk forecasting task against transcript-based risk prediction models. Risk in the financial market is often measured as the volatility of an underlying asset. There are different ways to measure volatility, and following Kogan, Levin, Routledge, Sagi, and Smith (2009), the risk of a focal stock between days t and $t + \tau$ is defined as the standard deviation of its stock's price return over a period of time:

$$v_{[t,t+\tau]} = ln\sqrt{\frac{1}{\tau - 1}\sum_{i=0}^{\tau}\left(r_{t+i} - \bar{r}\right)^2}, \tag{17}$$

where $r_t = \frac{p_t}{p_t - 1} - 1$ is the return of a stock with price p_t on day t, and $\bar{r}$ is the average return during the period. As a standard practice in finance, we take log since the log of volatilities tends to follow a bell distribution. A stock with high volatility indicates that its stock price fluctuates widely and thus is a high risk for investments, while a stock whose price stays more or less constant indicates a low risk for investments.

We consider two types of transcript-based risk forecasting methods as baselines. The first type includes methods specifically designed for transcript-based risk prediction. The second type involves using unsupervised document summarization techniques to extract key sentences, which are then fed into a pre-trained language model to generate transcript representations. These representations are subsequently input into MLP for risk prediction. For instance, in the case of ECT-SKIE, the set of key sentences $\kappa_\psi(X)$ extracted by ECT-SKIE is first encoded by BERT. The resulting dense vector (obtained through average pooling) is then fed into MLP to forecast risk.

$$\hat{y}_m = \text{MLP}(h_\theta(\kappa_\psi(X))). \tag{18}$$

In our experiments, $h_\theta(\cdot)$ is the pre-trained BERT. We use three-layer MLPs where the hidden size of each layer is set to 300, 150, and 50. In this experiment, all summarization models are set to extract 70% of the sentences from the transcript. We choose 70% because it strikes a balance between conciseness and comprehensiveness. A very low ratio, such as 10%, might be overly restrictive and risk omitting nuanced information that is crucial for accurate risk assessment. In addition, various risk prediction methods (Li, Yang, Smyth, & Dong, 2020; Qin & Yang, 2019; Sawhney et al., 2020; Yang, Ng, Smyth & Dong, 2020; Yang et al., 2023) utilize multimodal data, such as video and audio, for risk prediction. In our experiment, we only compare our method with those transcript-only methods.

- **Transformer-based (BERT)** (Koval et al., 2023)**:** Transformer-based language models have significantly improved text representations. While prior work has not applied transformer-based language models for the risk forecasting task, they have been used in forecasting earnings surprises – a relevant task (Koval et al., 2023). Thus, we implement a baseline using the transformer-based BERT model (Devlin, Chang, Lee, & Toutanova, 2019). Specifically, we feed each sentence in a transcript into the BERT model and then use mean-pooling to obtain the transcript-level representation. The representations are then fed into a linear layer to predict the associated risk.
- **TextRank** (Mihalcea & Tarau, 2004): TextRank is an unsupervised graph-based algorithm for natural language processing tasks such as keyword extraction and text summarization, where sentences or words are represented as nodes, and edges represent their semantic relationships, with the importance of each node determined by a variation of the PageRank algorithm.
- **LexRank** (Erkan & Radev, 2004): LexRank is an unsupervised graph-based method for text summarization that uses the concept of eigenvector centrality in a graph of sentences, where sentences are nodes connected by edges weighted by their cosine similarity, and more central sentences are considered more important for summarization.
- **PMI** (Padmakumar & He, 2021): This work presents an unsupervised summarization method that utilizes GPT-2 to calculate the pointwise mutual information between sentences. Note that PMI is quite time-consuming, requiring approximately 10 min to process a single transcript.
- **GenAI-based (GPT-4):** This method utilizes the cutting-edge GPT-4 model to generate abstractive summaries of earnings transcripts, which are then encoded using a pre-trained BERT model. We prompt GPT-4 with the following instruction: "Assume you are a financial analyst and are provided with the text of an earnings call transcript. Please extract the key sentences that are important for analyzing the company's performance. Output only key sentences, do not include any other reply".
- **Profet** (Theil, Broscheit, & Stuckenschmidt, 2019)**:** This work divides the earnings call transcripts into the presentation section, question section, and answer section, and then extracts section features for each section with BiLSTM and attention mechanism. Note that the simple division of the transcript into the three big sections is still quite coarse-grained.
- **MR-QA** (Ye et al., 2020)**:** This work designs a multi-round Q&A attention network, which represents the first attempt to decompose the large transcript into fine-grained Q&A level. MR-QA extracts features of each round of conversation through a bidirectional attention mechanism. Note that the between-QA and Presentation-QA structural information is ignored.

Table 2
Performance comparisons on the risk prediction task in terms of MSE and MAE. The best performance is in bold and the second-best results are underlined. Results are statistically significant ($p < 0.01$) under paired t-test.

Type	Method	Ratio	Mean Square Error (MSE)						Mean Absolute Error (MAE)					
			3d	7d	10d	15d	20d	60d	3d	7d	10d	15d	20d	60d
LM +Summarization	Transformer-based	100%	0.7401	**0.3603**	0.3070	0.2650	0.2340	0.1826	0.6724	0.4766	0.4295	0.3981	0.3728	0.3207
	+LexRank	70%	0.7521	0.3904	0.3282	0.2832	0.2577	0.1979	0.6768	0.4879	0.4474	0.4157	0.3957	0.3389
	+TextRank	70%	0.7528	0.3858	0.3244	0.2810	0.2562	0.1973	0.6779	0.4852	0.4451	0.4135	0.3951	0.3379
	+PMI	70%	0.7492	0.3775	0.3188	0.2703	0.2490	0.1900	0.6754	0.4806	0.4412	0.4007	0.3885	0.3309
	+GenAI-based	–	0.7966	0.4159	0.3403	0.2817	0.2538	0.1903	0.7032	0.5121	0.4508	0.4074	0.3816	0.3271
Risk prediction	Profet	100%	0.8058	0.4233	0.3343	0.3272	0.2585	0.2130	0.6947	0.5141	0.4485	0.4544	0.3927	0.3574
	MR-QA	100%	0.7868	0.3998	0.3025	0.2561	0.2311	0.1792	0.7022	0.4920	**0.4259**	**0.3888**	**0.3699**	0.3170
	DialogueGAT	100%	0.7432	0.3824	0.3238	0.2663	0.2315	0.1813	0.6740	0.4831	0.4419	0.3975	0.3688	0.3182
	DialogueEncoder	100%	0.7611	0.3952	0.3390	0.2608	0.2377	0.1948	0.6906	0.4933	0.4424	0.4310	0.3812	0.3336
	ECT-SKIE	70%	**0.7222**	0.3630	**0.3014**	**0.2557**	**0.2307**	**0.1744**	**0.6616**	0.4711	0.4270	0.3926	0.3706	**0.3146**

- **DialogueGAT** (Sang & Bao, 2022a)**:** This method models the speakers and their utterances in earnings calls by a graph attention network to forecast financial risk.
- **DialogueEncoder** (Sang & Bao, 2022b)**:** This method proposes using a pre-trained model to encode transcript sentences and concatenate them with sentence position embeddings, utterance embeddings, speaker's role embeddings, and section embeddings (Presentation and Q&A) for transcript dialogue representation.

Since risk prediction is a regression task (risk is a numerical value), we consider the Mean Squared Error (MSE) and Mean Absolute Error (MAE) regression metrics for evaluation. For all baselines, we tune their hyperparameters with a systematic search by minimizing MSE on the validation set. The hyperparameters of ECT-SKIE are set as (bold indicates the final choice): learning rate: $\{0.1, 0.01, \mathbf{0.001}, 0.0001\}$, $\beta : \{0, 0.01, \mathbf{0.1}, 1, 10\}$, $K\{50, 100, 200, \mathbf{400}, 800\}$, $\alpha : \{0.1, 0.3, 0.5, \mathbf{0.7}, 0.9\}$. We fix batch size to 128, set the temperature of NCE to 0.1, and use the AdaGrad (Duchi, Hazan, & Singer, 2011) for model training.

The experimental results of risk prediction are presented in Table 2. When examining the performance of baselines specifically designed for risk prediction, we have the following findings. Firstly, a fine-grained decomposition of the transcript consistently leads to better risk prediction performance compared to coarse-grained approaches. For example, Profet, which simply divides the earnings call transcript into three sections, underperforms compared to the other three baselines that consider interactions among Q&A rounds, individual utterances, and speakers. Secondly, the Q&A section proves to be crucial for accurate risk prediction. As shown, MR-QA, which focuses exclusively on decomposing the transcript at the fine-grained Q&A level, achieves superior results compared to methods that model speaker information. This observation further validates our design rationale, where Q&A sections are used to create a surrogate supervision signal for implementing self-supervised learning. Thirdly, with an information compression ratio of 70%, ECT-SKIE significantly reduces forecasting errors across different time intervals. For instance, the mean squared error (MSE) of predicted risk 60 days after the earnings call is 0.1744, even outperforming models specifically designed for the risk prediction task. It also suggests that filtering out irrelevant noise, such as boilerplate sentences, can lead to higher-quality transcript representations and improve prediction performance.

When examining the performance of baselines related to summarization, we have the following findings. Firstly, all summarization methods, including TexRank, LexRank, and PMI, underperform compared to the uncompressed base model, which means that they fail to extract all key sentences related to risk prediction. A major reason for this is that these general-purpose algorithms are not specifically designed to capture the nuances of financial text. They primarily rely on generic sentence ranking techniques, which overlook domain-specific signals crucial for risk prediction, such as the financial implications embedded in the Q&A interactions and the sentiment or emphasis conveyed in the wording of key statements. Secondly, even GPT-4, which generates abstractive summarizations, performs worse than traditional extraction-based approaches. This may be due to its tendency to generalize, potentially losing vital details that are key to financial decision-making (Zhou et al., 2023). Furthermore, GPT-4 generate summaries rather than directly extracting key sentences from the original text, leading to a lack of interpretability compared to other extractive methods. Finally, our method is the only one that surpasses the base model, confirming the importance of tailoring extraction techniques to the specific characteristics of financial data, rather than relying on generic summarization methods.

5.3. Experiment 2: Overall Performance of ECT-SKIE on Transcript Retrieval

Transcript retrieval aims to find relevant documents in a collection given the query (Guo et al., 2022). This process can be used to evaluate the ability of ECT-SKIE and other unsupervised extractive summarization methods to extract discriminative information from earnings call transcripts. Given a list of earnings call transcripts $d_1, d_2, \ldots, d_n$, we consider the extracted sentences as a query q and rank these candidate transcripts based on their relevance to the query q, which can be formalized as a probability distribution over transcripts:

$$p(d_i|q) = \frac{\exp\left(\mathrm{sim}(\phi(q), \phi(d_i))\right)}{\sum_j \exp\left(\mathrm{sim}(\phi(q), \phi(d_j))\right)} \tag{19}$$

Table 3
Performance comparisons on transcript retrieval task. We calculate computational time on the same device.

Metric	LexRank	TextRank	PMI	GenAI-based (GPT-4)	ECT-SKIE
Precision	0.6400	0.8380	0.9300	0.8946	**0.9440**
Mean rank	4.6800	2.4020	2.1020	2.2875	**1.7100**
Mean reciprocal rank	0.7274	0.8821	0.9595	0.9271	**0.9627**
Seconds per sample	0.2362	0.2727	1.6662	–	**0.0039**

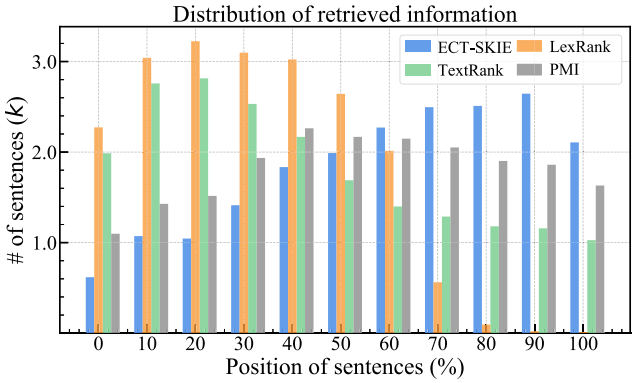

Fig. 4. The percentage position distribution of the extracted sentences in source transcripts.

where the $\mathrm{sim}(\cdot,\cdot)$ is the cosine similarity function and the vector encoder $\phi(\cdot)$ is a pre-trained BERT model. We evaluate the overall performance of the proposed method ECT-SKIE on the transcript retrieval task against three extractive summarization approaches TextRank, LexRank, PMI and one abstractive summarization approach GenAI-based (GPT-4). For extractive methods, we set the information compression ratio as 15%. For abstractive method (GPT-4), we use the same prompt in the risk prediction task. We use three widely used metrics: Precision, Mean Rank (i.e., the mean position of the true transcript after ranking all transcripts based on $p(d_i|q)$), and mean reciprocal rank (MRR) (i.e., the mean of the inverse of the rank) for performance evaluation.

Table 3 validates ECT-SKIE's capability in extracting the key insights. Combined with the statistics described in Fig. 4, it is clear that achieving superior retrieval performance is closely related to the capabilities of extracting insights from Q&A section. The distribution of extracted sentences reveals that ECT-SKIE allocates greater attention to the Q&A section. This trend can be partially attributed to the Q&A-based auxiliary supervision signals that guide the ECT-SKIE's learning process. In contrast, TextRank and LexRank show a tendency to emphasize sentences that appear early in the transcript, mainly in the presentation section. GPT-4 also pays more attention on the presentation section than the Q&A section when asked about its focus. Generally, the sentences early in an earnings call transcript often reflect a firm's overall business situation as described by its management. These sections typically contain prepared statements or highlights of recent performance. In contrast, sentences later in the transcript, particularly those in the Q&A section, tend to focus on more specific concerns raised by analysts and investors, revealing deeper insights into potential risks or challenges. This suggests that the Q&A section often contains more detailed and critical information, which may provide unique insights that help differentiate one company's outlook from another. It is worth noting that ECT-SKIE extracts key insights in parallel, resulting in a dramatic speedup (~500 times) compared to other methods.

5.4. Experiment 3: Qualitative Analysis of ECT-SKIE on Firm Discovery

In this experiment, we conduct a firm discovery experiment to better understand the quality of learned transcript representations. Specifically, we randomly select four companies from the testing set as the focal company, and calculate the cosine similarity between the focal company's transcript embeddings and other company's transcript embeddings. Intuitively speaking, companies with similar products and services should have similar transcript representations. Prior business research (Hoberg & Phillips, 2016) has empirically shown that companies' corporate disclosures can be used to discover economically related firms, which serves as the basis of text-based market analysis.

We select the ten most similar companies to the focal company, and present the results in Table 4. The results show that the transcript embeddings learned by our approach can identify economically related companies. For example, the most similar companies to Visa Inc. are Mastercard Inc. and Citigroup Inc. Similarly, Alibaba is a well-known company in the e-commerce sector, and its similar companies include Pinduoduo Inc. -ADR, Shopify Inc., ChannelAdvisor Corp., and Groupon Inc., all of which are in the e-commerce sector. In summary, the firm similarity experiment provides face validity that our approach is capable of learning semantically meaningful transcript representations. This also implies that our approach can be utilized, beyond risk predictions, in a wide range of financial tasks such as market competition analysis.

Table 4
Top ten most similar firms to focal firms.

Focal firm	Tesla Inc.	Alibaba Group HLDG	Intel Corp.	Visa Inc.
Similar firms	Delphi Technologies PLC	Pinduoduo Inc. -ADR	Advanced Micro Devices	Mastercard Inc.
	Methode Electronics Inc	Shopify Inc.	Axcelis Technologies Inc.	Citigroup Inc.
	Enphase Energy Inc.	ChannelAdvisor Corp.	Applied Materials Inc.	Evertec Inc.
	Polar Power Inc.	Groupon Inc.	NXP Semiconductors NV	Banco Santander SA
	Constellium SE	Wayfair Inc.	Integrated Device Technology Inc.	Northern Trust Corp.
	Navistar, Inc.	Criteo S.A.	IBM Corp.	Euronet Worldwide, Inc.
	Penske Automotive Group, Inc.	Leju Holdings Ltd	Micron Technology, Inc.	EVO Payments, Inc.
	Ferrari N.V.	Yiren Digital Ltd.	NVIDIA Corp.	PayPal Holdings, Inc.
	FreightCar America, Inc.	Vipshop Holdings Ltd.	Synopsys, Inc.	Cardtronics PLC
	UQM Technologies, Inc.	TechTarget, Inc.	Pure Storage, Inc.	Western Union Co.

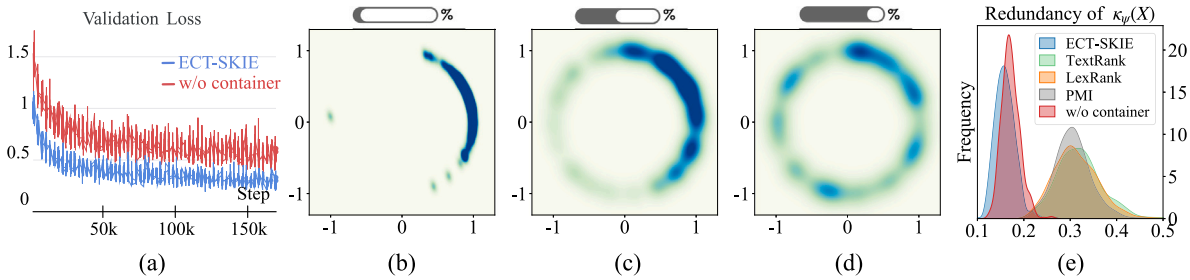

Fig. 5. (a) The loss curves of the container ablation. (b, c, d) The three pictures show feature distributions of slots in containers at different training periods, derived by Gaussian kernel density estimation (KDE) in $\mathbb{R}^2$. (e) The redundancy distribution computed by cosine similarity between extracted sentences.

5.5. Redundancy investigation

As discussed in Section 4.1, the issue of high redundancy significantly impacts the efficiency of information compression. To substantiate the effectiveness of our proposed redundancy-aware key sentence extractor, we conducted two meticulous experiments, tracking the changes in training loss curves (Fig. 5(a)) and the evolution of feature distribution throughout the training process (Fig. 5(b–e)).

As illustrated in Fig. 5(a), the model equipped with the container exhibits a notably lower training loss curve compared to its vanilla counterpart. This intriguing phenomenon aligns with our hypothesis that the container effectively addresses the redundancy issue. Fig. 5(b–d) shows that as training continues, the distribution of slots and their corresponding features gradually becomes more diverse. Each slot within the container can be interpreted as representing different financial risk factors. As training proceeds, the container mechanism enables the model to comprehensively consider a range of risk factors, thereby enhancing the understanding derived from the earnings call transcript.

We further quantitatively measure the redundancy of the extracted key sentences. For the key sentences extracted from one transcript, the redundancy score, Red, is computed as follows:

$$Red = 2\left(\sum_{j=1}^{N_s-1}\sum_{k=j+1}^{N_s}\cos(\mathbf{x}_j,\mathbf{x}_k)\right)/(N_s^2 - N_s). \tag{20}$$

A lower value of Red indicates less redundancy. Fig. 5(e) plots the redundancy score distribution of four baselines, and we can see that ECT-SKIE yields lower sentence redundancy compared to its variant (i.e., w/o container) and significantly better than three baselines.

5.6. Sensitivity analysis of K and subjective evaluation

Since K is the most important hyper-parameter in ECT-SKIE, we investigate the sensitivity of the container size and present the results in Fig. 6. We can see that the loss decreases as K increases, which is expected. As we discussed in Section 4.1, each slot in the container matches with one candidate sentence, and one candidate sentence might match with multiple slots. The larger K means there are more slots in the container, and the lower the chance that a candidate sentence matches multiple slots. Therefore, when K is set to a larger value, it helps to generate a more comprehensive understanding of the transcript.

We conduct two subjective evaluation experiments. The first evaluates the quality of the sentences extracted by ECT-SKIE, and the second compares the subjective quality of ECT-SKIE with PMI. Ten volunteers, all MS/PhD candidates from business or computer science schools with specializations in risk management, venture capital, and artificial intelligence, are invited to participate. Volunteers are asked to rate the extracted sentences based on their informativeness and relevance to financial risk. In Table 5, the preference ratios indicate that our model outperforms PMI.

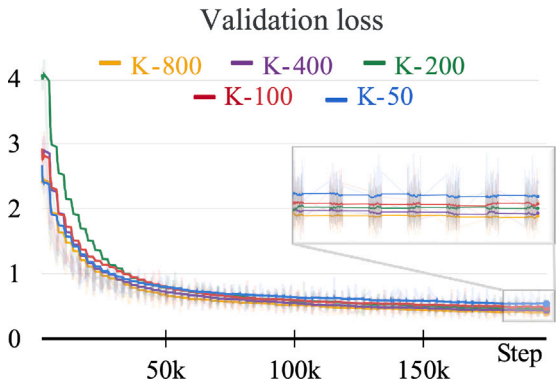

Fig. 6. Loss curves of different container size K on the validation set ($\alpha = 0.15$).

Table 5
Subjective evaluation for the quality of extracted sentences by 10 volunteers.

Volunteers	Case										Score	Preference (%)	
(Number)	1	2	3	4	5	6	7	8	9	10	(mean)	PMI	ECT-SKIE
No. 1	8.5	7.0	7.0	7.5	4.0	6.0	7.5	7.0	7.5	8.0	7.00	40%	60%
No. 2	9.0	7.0	7.0	8.0	5.0	5.0	8.0	8.0	7.0	8.0	7.20	30%	70%
No. 3	8.5	7.0	3.0	8.0	6.5	7.5	8.0	8.0	8.5	8.0	7.30	30%	70%
No. 4	6.5	5.0	8.0	3.0	4.0	6.0	8.0	6.0	6.0	8.0	6.05	60%	40%
No. 5	8.0	6.0	7.0	7.0	5.0	7.0	8.0	8.0	5.0	6.0	6.70	60%	40%
No. 6	7.0	6.5	7.0	8.0	5.0	7.0	8.0	7.0	4.0	5.5	6.50	40%	60%
No. 7	6.5	7.0	7.0	8.0	6.0	7.5	7.0	8.0	7.0	6.0	7.00	20%	80%
No. 8	8.0	6.0	6.5	6.0	4.0	6.0	8.0	8.0	8.0	8.0	6.85	40%	60%
No. 9	4.0	5.0	8.0	4.5	6.0	9.0	3.0	8.5	6.0	7.0	6.10	50%	50%
No. 10	7.0	4.0	6.0	8.0	6.5	5.0	8.0	7.0	7.0	6.0	6.45	20%	80%

5.7. Interpretability analysis

In this section, we conduct an interpretability analysis using the 2018 Fourth Quarter transcript of Varian Medical Systems Inc. as a case study.

In the presentation section, the company reports strong revenue growth, particularly highlighting the robust performance of the Halcyon system and regional growth across the Americas, EMEA, and Asia-Pacific. During the Q&A section, analysts raise questions about Halcyon order growth, tariff impacts, and the sustainability of market growth. For instance, analyst Amit Hazan inquires about US–China tariffs, a topic that appears in both sections, indicating its significant relevance to the company's performance.

Our experiments show that PMI, GPT-4 and ECT-SKIE can accurately extract specific information regarding US–China tariffs from the presentation section, such as the statement, "The US–China tariffs impacted our Q4 results". However, ECT-SKIE captures more detailed insights from the Q&A section. For example, it retrieves the sentence, "Through the operating actions we talked about in terms of supply chain, commercial operations, G&A productivity, and our R&D rephrasing, we can net that back down to an $18 million impact", providing deeper context on the tariff issue. In contrast, all baselines overlook this detail, and GPT-4 even outputs truncated sentences, resulting in the loss of some semantic information and causing ambiguity. This highlights the distinct advantage of our method in extracting key insights that are critical for understanding risk factors.

6. Discussions on implications

We discuss the implications for practitioners and researchers.

Theoretical implications. This study contributes to the theoretical framework of text analysis by introducing ECT-SKIE, an innovative self-supervised method for key insight extraction. By employing variational information bottleneck theory, we advance the understanding of information retention and extraction processes in NLP, particularly in financial contexts. The integration of structure-aware contrastive learning further enhances the theoretical discourse surrounding self-supervised learning paradigms, emphasizing the significance of context and structure in effective text summarization. Moreover, the proposed container-based key sentence extractor designed to address the sentence redundancy issue enriches the theoretical implications of managing redundancy in extractive summarization methods. Overall, this research lays the groundwork for future investigations into the intersection of machine learning, natural language processing, and financial analysis.

Practical implications. Earnings conference calls serve as a crucial medium for corporate communication, delivering insights and disclosures that may not yet be reflected in financial reports (Li, 2010). For market participants, including portfolio managers,

investors, and financial analysts, our proposed method provides a refined and advanced method for analyzing the discussions that take place during these earnings calls (Kogan et al., 2009). Some significances include:

- **Enhanced Decision-Making and Operational Efficiency**: By automating the extraction of key insights from transcripts, this method significantly improves decision-making for investors and analysts (Bushee & Miller, 2012). Traditionally, professionals spent hours on manual analysis. Our approach streamlines this process, allowing them to focus on strategic analysis and reducing human error. Additionally, it accurately predicts financial risk, enhancing risk management (Froot, Scharfstein, & Stein, 1993). The insights extracted by our model are more objective and avoid subjective influence, improving investment decisions and portfolio management.
- **Competitive Analysis and Market Research**: The tool identifies economically related firms through transcript analysis, which is useful for competitive analysis and market research. Companies can monitor competitors' strategies and gain insights into market trends, informing strategic planning (Zuckerman, 2000). Analyzing competitors' calls provides valuable intelligence on industry trends and growth opportunities.
- **Generalization and Scalability to Other Financial Text Processing:** ECT-SKIE's design principles and methodologies exhibit strong scalability to other financial text processing tasks, such as the analysis of annual financial statements, regulatory filings, and market reports. The core components – variational information bottleneck theory and structure-aware contrastive learning – are adaptable to various financial documents that require extracting key insights from large text volumes. For instance, annual financial statements and regulatory filings like 10-K and 10-Q reports can benefit from ECT-SKIE's ability to identify and summarize critical information efficiently. By leveraging the model's self-supervised learning capabilities, financial analysts can automate the extraction of valuable insights across different document types, enhancing the accuracy and efficiency of financial analysis and decision-making processes. This scalability underscores ECT-SKIE's versatility and potential as a foundational tool in financial text analysis.

To maximize the model's impact, we have made the code publicly available, allowing practitioners to reproduce the results, build upon the findings, and implement them in subsequent financial applications to foster the development of more advanced and effective financial analysis tools.

7. Concluding remarks

In this study, we proposed ECT-SKIE, an efficient self-supervised approach for extracting key insights from earnings call transcripts. We introduced variational information bottleneck technology to extract key insights in parallel, significantly speeding up the extraction process. Additionally, we devised a structure-aware contrastive learning method, enabling model training without reliance on any labeled data. We further proposed a novel container-based key sentence extractor to minimize redundancy. Experimental results on a large-scale dataset demonstrate the effectiveness of the proposed approach for key sentence extraction. Given the prevalence of lengthy text documents in the financial domain, we believe our implementation can benefit researchers and practitioners on other text-based financial analysis tasks.

Our work has some limitations. First, each container slot is matched with one candidate sentence, which means that one sentence could correlate with multiple slots, potentially affecting the consistency of the extracted sentences with the desired compression ratio. Second, the compression ratio in our model is fixed, lacking the flexibility to adapt to each earnings call transcript dynamically. We will try to address the above limitations in future work.

CRediT authorship contribution statement

Yanlong Huang: Writing – original draft, Software, Methodology. **Wenxin Tai:** Writing – review & editing, Methodology. **Fan Zhou:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Qiang Gao:** Writing – review & editing, Investigation. **Ting Zhong:** Supervision, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used GPT-4 to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the submitted work.

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Data availability

Data will be made available on request.

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